```
    centroids ← mapReduce(T1, T2, mappers, reducers)
                  #6
    if |centroids| == numCentroids then
                  #7
      break
    elsif |centroids| > numCentroids
then                                          #8
      delta ←
random(T2)
#9
      T2 ← T2 +
delta
#10
      T1 ← T1 + delta
                                                    #11
    else
      T2 ← T2 –
random(T2)
#12
  return addRandomNoise(centroids)
                                              #13
```

#1 Method `MRcanopyCentroids` takes a list of points, the number of desired centroids, and two thresholds, `T1` and `T2`, to be used for canopy clustering initialization step. It also takes the number of shards

#2 Break down the dataset into `numShards` random shards. This is usually done automatically by the master node of map reduce, but in this case, we will use a customized version.

#3 Initialize `numShards` mappers, each will run the `canopyClustering` method and will hold a copy of one of the dataset's shards. The mappers will be alive (and hold the same copy of the input) for the whole lifetime of this method.